



Skyline Stats

U.S. Domestic Flights Performance Analysis (2015)

A Data-Driven Approach to Identifying Delay and Cancellation Drivers

Track

Data Analytics

Submitted To
Digital Egypt Pioneers Initiative (DEPI)

Version 1.0



Problem Statement :



Airline delays and cancellations significantly affect operational efficiency and passenger satisfaction.

With increasing air traffic, identifying performance bottlenecks has become more critical

The aviation industry lacks a clear analytical view of patterns behind disruptions such as peak times, hub congestion, and seasonal effects.

There is a need for a data-driven solution to support informed decision-making for airlines, airports, and regulators.

Project Goal: Actionable Insights for Aviation Stakeholders

Objective: Analyze 5.8M+ U.S. domestic flights from 2015 to uncover patterns in delays and cancellations, providing actionable insights for aviation industry stakeholders.

 Key Finding 1: Time-Based Disruptions

Evening hours (6 PM - 9 PM) and Thursdays experience the highest disruption rates, indicating predictable demand spikes.

 Key Finding 2: Hub Congestion

Major hubs like ATL and ORD handle the most traffic but also account for a high volume of cancellations.

 Key Finding 3: Strategic Opportunity

The analysis provides a strategic tool for improving operational efficiency and enabling data-driven decision-making.

5.8M+

Total Flights Analyzed

2015

Analysis Year

14

Airlines Covered

628

Departure Airports

Primary Deliverable: Interactive Power BI Dashboard with multi-page analysis and real-time filtering capabilities.

Structured Data Analysis Lifecycle

Objective: Analyze and visualize 2015 U.S. domestic flight data to uncover patterns in delays and cancellations, providing actionable insights for aviation industry stakeholders.

IN-SCOPE

- 2015 U.S. domestic flights
- Airline performance
- Airport performance
- Route analysis
- Temporal patterns
- Interactive dashboard

OUT-OF-SCOPE

- International flights
- Ticket pricing analysis
- Economic impact
- Passenger satisfaction
- Revenue forecasting

METHODOLOGY: 5-STEP APPROACH

- 1

Data Gathering

Aggregated flight, airport, and airline datasets from Kaggle (5.8M+ records)
- 2

Data Preparation

Cleaning, transformation, and zero-imputation for missing delay values using Power Query
- 3

Data Modeling

Established Star Schema with fact and dimension tables, created DAX measures
- 4

Visualization

Developed multi-page interactive Power BI dashboard with cross-filtering
- 5

Insight Generation

Extracted and documented key analytical findings with actionable recommendations

Tools & Technologies Used

Power BI:

**Data modeling, dashboards, and
visualization**

Power Query:

Data cleaning and transformation

DAX:

**Calculated measures (Cancellation Rate,
On-Time Performance, performance
metrics)**

CSV Dataset (Kaggle):

5.8M+ U.S. domestic flights (2015)

Robust Star Schema for Flexible Analysis

DATA SOURCES

flights.csv
Transactional data for each flight (5.8M+ records)
airports.csv
Dimension table with airport details (IATA codes, names, locations)
airlines.csv
Dimension table with airline names and IATA codes

DATA PREPARATION HIGHLIGHTS

- ✓ **Missing Value Handling:**
Imputed missing delay values with zero, as missing indicates no recorded delay
- ✓ **Data Type Validation:**
Ensured correct data types for numerical and date fields
- ✓ **DAX Measures:**
Created key measures: Cancellation Rate, On-Time Performance

STAR SCHEMA STRUCTURE

Fact Table
flights (5.8M+ records with delay, cancellation, and route data)
Dimension Tables
airports • airlines

POWER BI RELATIONSHIPS

Manage relationships

✕

+ New relationship

 Autodetect

 Edit

 Delete

 Filter ▾

☐	From: table (column) 	Relationship	To: table (column)	Status	
☐	flights (AIRLINE)		airlines (IATA_CODE)	Active	⋮
☐	flights (DESTINATION_AIRPORT)		airports (IATA_CODE)	Inactive	⋮
☐	flights (ORIGIN_AIRPORT)		airports (IATA_CODE)	Active	⋮

KEY RELATIONSHIPS

- Flights [AIRLINE] → Airlines [IATA_CODE] (Active)
 - Flights [ORIGIN_AIRPORT] → Airports [IATA_CODE] (Active)
 - Flights [DESTINATION_AIRPORT] → Airports [IATA_CODE] (Inactive)
- The destination airport relationship is intentionally inactive to prevent ambiguity and enable dual analysis via DAX.

Tailored Insights for Key Aviation Stakeholders



Airlines

Business Questions

- Which routes are underperforming?
- How do we compare to competitors on key delay metrics?
- What operational improvements are needed?
- Which aircraft types have higher delays?



Airports

Business Questions

- What are peak hours for delays?
- Which airlines cause the most delays?
- How should we allocate resources?
- What is our operational efficiency?



Passengers

Business Questions

- Which airline should I choose?
- What time of day is best to travel?
- Which day of week has fewer delays?
- How reliable are specific routes?



Regulators

Business Questions

- What are systemic causes of delays?
- How is the national air system performing?
- Which airlines need intervention?
- What trends are emerging?

User-Centric Design for Intuitive Data Storytelling

Design Philosophy: The dashboard was designed with a user-centric approach, focusing on clarity, simplicity, and intuitive navigation. The color scheme was chosen strategically to guide user attention and facilitate rapid insight discovery.

Color Coding Strategy

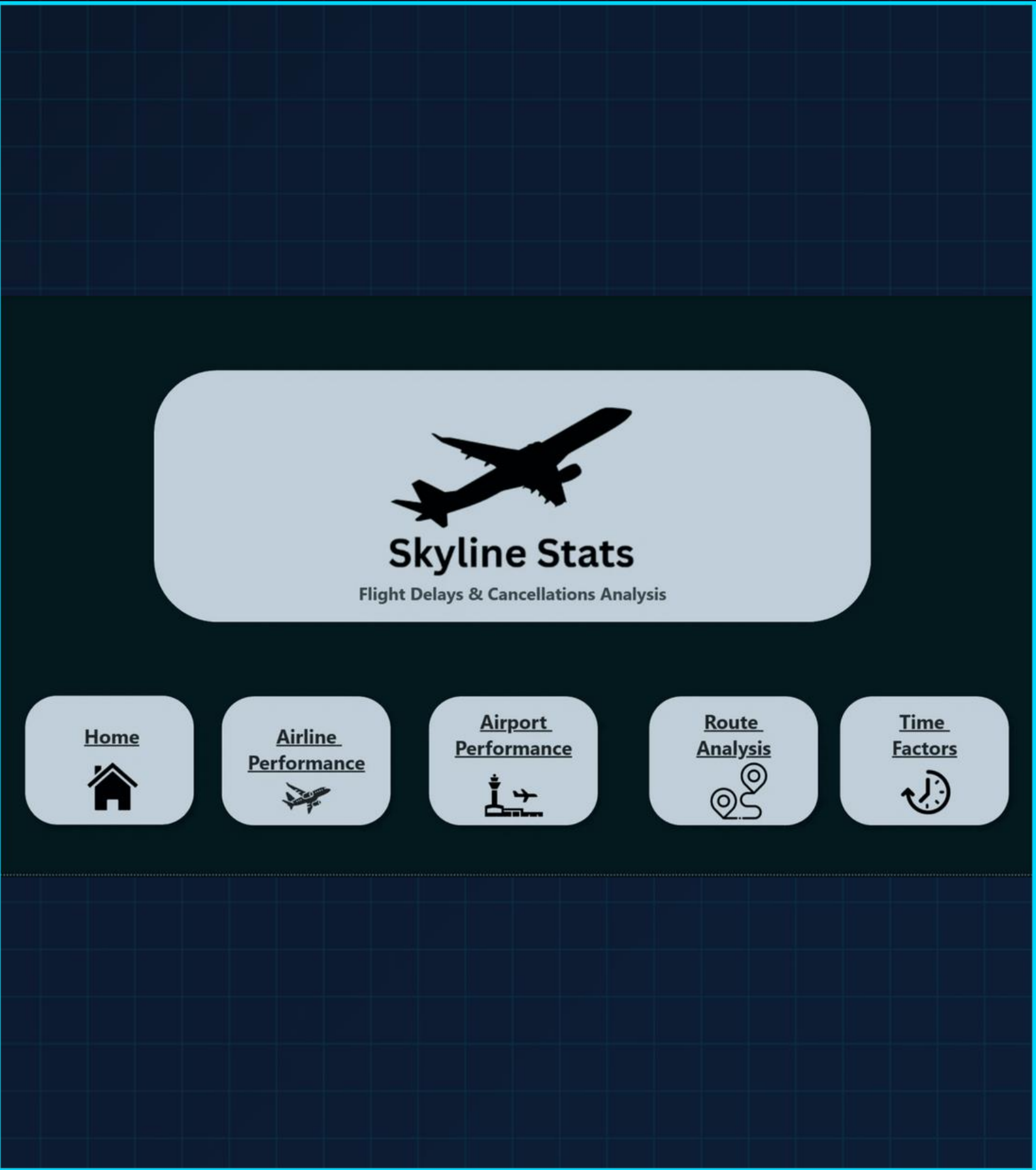
Red and orange tones highlight negative metrics (delays, cancellations). Neutral colors provide context. This "data storytelling" approach allows users to grasp key takeaways at a glance.

Multi-Page Architecture

Each page is dedicated to a specific analytical theme, enabling focused exploration while maintaining a cohesive visual identity across all sections.

DASHBOARD STRUCTURE

- **Home Page:** Navigation hub with high-level overview and buttons to access detailed sections.
- **Airline Performance:** Deep dive into carrier-specific metrics and comparisons.
- **Airport Performance:** Analysis of hub and regional airport metrics.
- **Route Analysis:** Route-specific performance and patterns.
- **Time Factors:** Temporal patterns and seasonal trends.



Operational Efficiency Varies Significantly Across Carriers

✈️ ON-TIME PERFORMANCE: LEGACY vs LOW-COST

Legacy Carriers (DL, AS): Demonstrate superior on-time performance with lower average delays, indicating more mature operational processes and resource management.

⚠️ HIGH DELAY RISK: LOW-COST CARRIERS

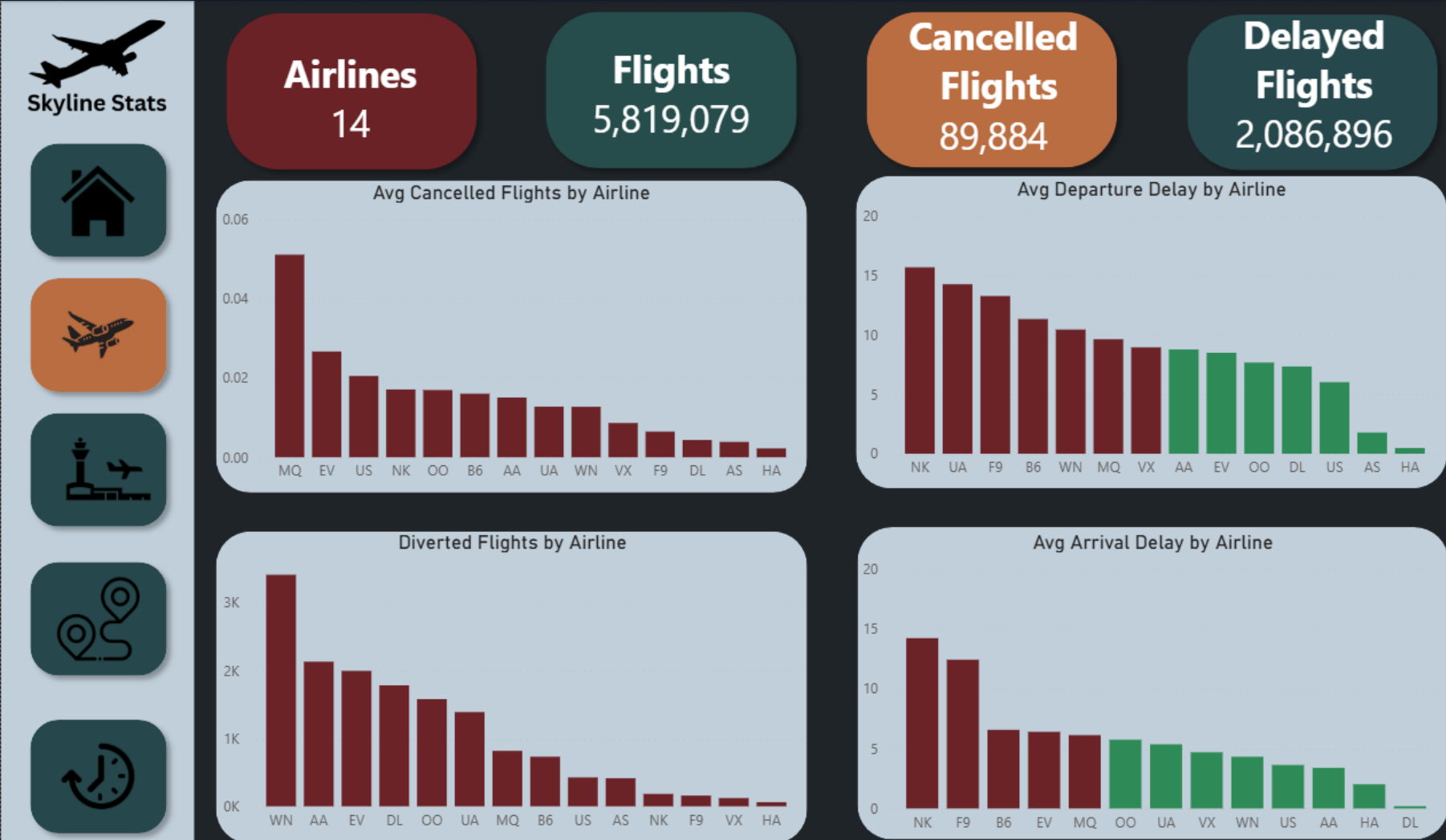
Low-Cost Carriers (NK, F9): Exhibit significantly higher average delays, suggesting operational constraints or aggressive scheduling that impacts punctuality.

🚨 CRITICAL FINDING: AMERICAN EAGLE (MQ)

American Eagle has the **highest cancellation rate** among all carriers, indicating potential operational or fleet-specific issues that require immediate attention and investigation.

💡 STRATEGIC IMPLICATION

Airlines with high delay rates should review their turnaround schedules and contingency plans for high-traffic routes to improve operational resilience.



Major Hubs Drive High Volume of Cancellations

High Traffic & Cancellations

Major hubs like Chicago O'Hare (ORD), Dallas/Fort Worth (DFW), and Hartsfield-Jackson Atlanta (ATL) record the highest number of total cancellations, largely due to their high volume of flights.

Top Airports by Cancellation Volume

ATL - Hartsfield-Jackson Atlanta

Highest cancellation volume due to busiest airport status

ORD - Chicago O'Hare

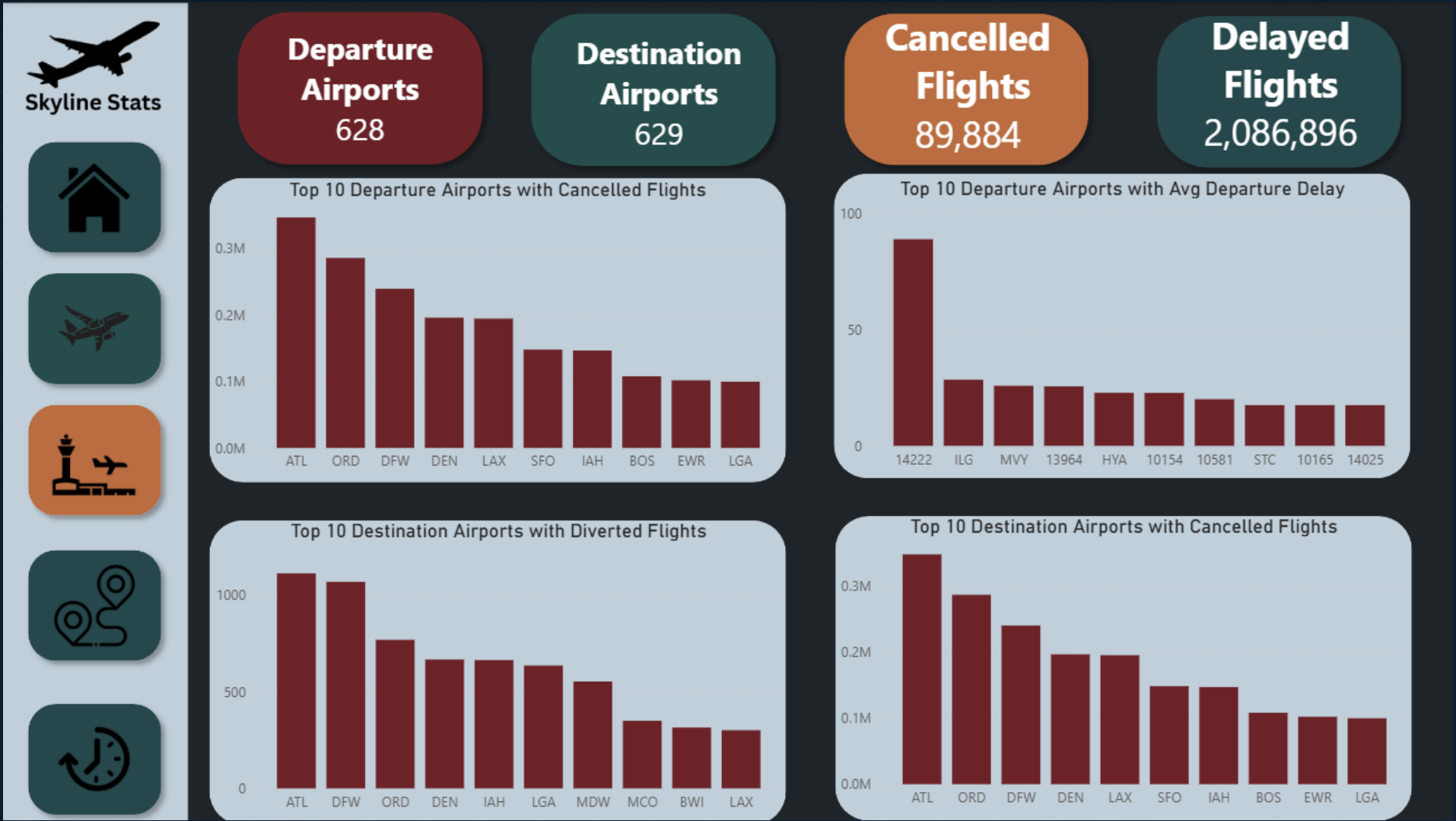
Second highest volume; major hub for connecting flights

DFW - Dallas/Fort Worth

Third highest; significant hub for American Airlines

Regional Airport Vulnerability

Smaller, regional airports sometimes show high average delay times, suggesting they are more susceptible to network disruptions originating from larger hubs.



Route Analysis

High Volume of Flights

West Coast Shuttle: The connection between **Los Angeles (LAX)** and **San Francisco (SFO)** is the single most critical artery in this network. It appears as the #1 and #2 top routes.

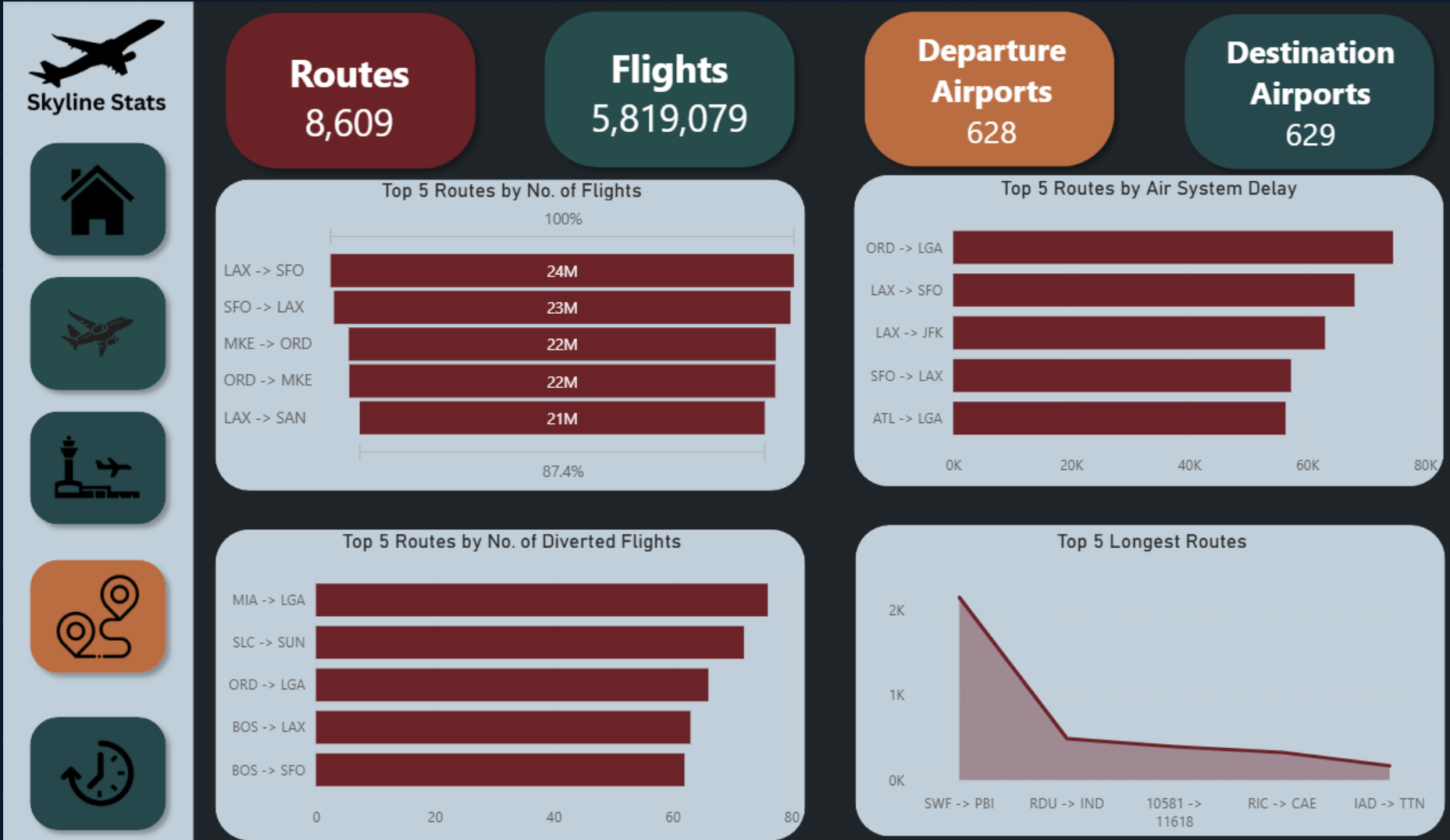
The Risky Routes

Weather/Mountain Issues: The route **SLC (Salt Lake City) -> SUN (Sun Valley)** has a high number of diversions. Sun Valley is a mountainous airport known for difficult weather conditions, making this a "high risk" route for completing the journey as planned

Miami to New York: **MIA -> LGA** is the #1 route for diversions. This is likely due to the combination of Florida's notoriously volatile weather (thunderstorms) and New York's tight airspace.

The Double Trouble

The **LAX -> SFO** and **SFO -> LAX** routes are high volume *and* high delay. This is a classic capacity issue where popularity is outstripping infrastructure capabilities.



Predictable Time-Based Demand Spikes

🕒 TIME OF DAY: EVENING PEAK

Delays accumulate throughout the day, **peaking in the evening (6 PM - 9 PM)**. This pattern is consistent across all days of the week and reflects increased traffic during evening commute and leisure travel hours.

📅 DAY OF WEEK: WEEKEND TRAVEL SURGE

Thursdays and Fridays experience the highest volume of delays, likely due to the start of weekend travel. This predictable pattern enables proactive resource planning.

❄️ SEASONALITY: WINTER WEATHER IMPACT

Winter months (December, January, February) show a significant spike in weather-related delays and cancellations, indicating seasonal vulnerability in the national air system.

💡 STRATEGIC OPPORTUNITY

Passengers: To minimize delay risk, book morning flights and avoid travel on Thursdays/Fridays when possible. **Airlines & Airports:** Focus resource allocation during peak evening hours and weekend travel periods.

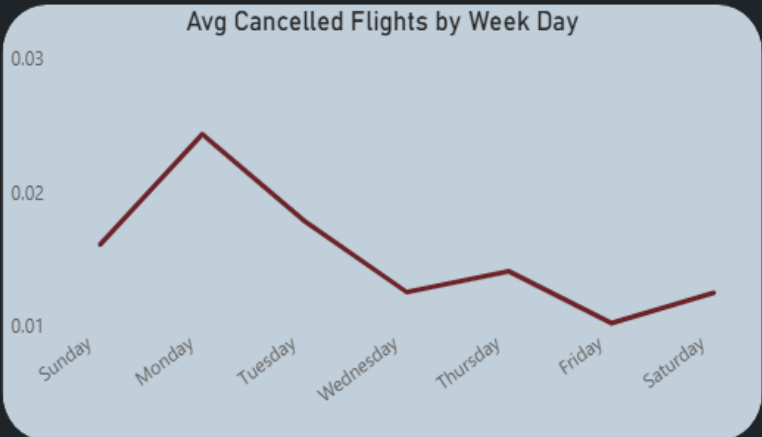
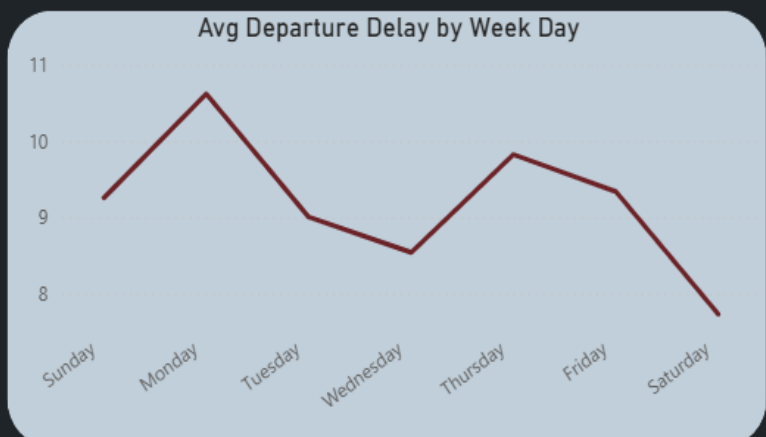
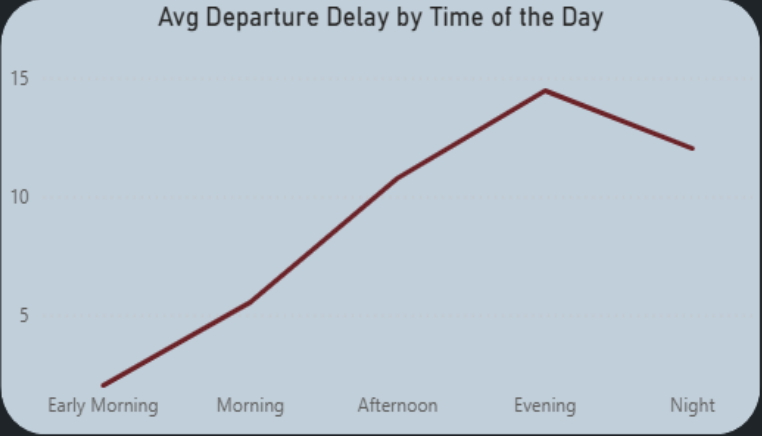
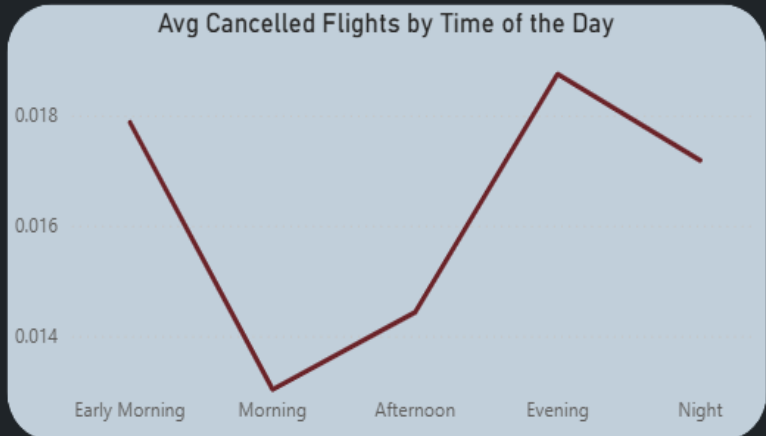


Time With The Most Flights Morning

Day With The Most Flights Thursday

Avg Departure Delay 9.23 min

Avg Arrival Delay 4.33 min



Strategic Recommendations for Improved Operational Resilience

CONCLUSION: PRIMARY INEFFICIENCY DRIVERS

Analysis of 2015 U.S. flights identified key delay/cancellation patterns. Inefficiency stems from **airline operations, airport congestion, and predictable time-based spikes**—interconnected issues requiring coordinated action.

FOR AIRLINES

Strengthen resilience for evening flights; review turnarounds and contingencies on high-traffic routes, and apply dynamic scheduling in peak periods.

FOR AIRPORTS

Allocate more resources during peak evenings and Thu/Fri; invest in de-icing/runway tech where weather delays are high; use predictive staffing.

FOR PASSENGERS

Book morning flights and avoid Thursdays when possible; use delay patterns to choose travel times and carriers.

FOR REGULATORS

Track systemic trends and promote best practices; inform capacity planning and operational standards with these insights.





You can view our interactive Power BI dashboard using the link attached below:

https://hnueg-my.sharepoint.com/:u:/g/personal/hana_ali_2240203_cit_hnu_edu_eg/IQDHNEJxOo3zS4XjYLdY2eVTAR8KP3izeQHahSw0I41W98s?e=9KiVED

Thank You

Questions & Discussion

We welcome your insights, questions, and feedback on our analysis of U.S. domestic flight performance.

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